

BASICS FOR AIR TRAFFIC CONTROL – HAZARDOUS WEATHER

INTRODUCTION

Your job as an air traffic controller is to provide a service to the flying public. Weather is perhaps the most significant factor that affects the flow of air traffic and accounts for a significant percentage of all accidents. FAA Order JO 7110.65 states that controllers shall advise pilots of hazardous weather that may impact operations within 150 nautical miles (NM) of their sector or area of jurisdiction. Therefore, it is critical that you are able to identify the characteristics of hazardous weather and their effects on aircraft.

This module discusses the characteristics of hazardous weather and their effects on aviation.

The hazards covered in this module each have their own distinct and very real dangers for pilots. You will be able to identify these characteristics by the end of this module.



ADVERSE WEATHER

Purpose: This lesson identifies the severity and impact that weather hazards and adverse wind have on aircraft, especially during takeoff and landing.

Objectives:

- Define adverse weather conditions that can affect an aircraft
- Define adverse wind effects that affect aircraft

References for this lesson are as follows:

- Aeronautical Information Manual (AIM)
- American Meteorology Society (AMS), Glossary of Meteorology
- UCAR/COMET: Writing TAFS for Winds and LLWS
- NASDAC Review of NTSB Weather Related Accidents

Aviation Weather Hazards

An **aviation weather hazard** is an **atmospheric condition** that, when encountered in flight, can potentially cause damage to the aircraft, personal injury, a crash, or death.

Pilot and aircraft capabilities are factors that must be considered.

- Some hazards impact all flights
- Some hazards only impact pilots and aircraft with limited capabilities

Factors that influence aviation weather safety:

Pilot	Ratings and Experience ■ Instrument rated ■ Visual Flight Rules (VFR) only
Aircraft	Design Performance Specifications ■ Power ■ Speed ■ Range ■ Service ceiling
Equipment	Onboard Equipment ■ De-ice/anti-ice ■ Navigational aids ■ Autopilot ■ Radar ■ Lightning detector (Stormscope or Strikefinder) ■ Uplink weather

Adverse Wind Effects

Adverse wind is responsible for most weather-related accidents and often triggers air traffic management decisions that adversely impact traffic.



Adverse wind's impact on:

Aircraft	■ General aviation (GA) pilots flying aircraft with lower crosswind and tailwind threshold values are the most at risk ■ Takeoff and landing are the most critical phrases of flight and are most susceptible to adverse wind
Air Traffic Control	■ Change of runway configuration – Wind determines which direction planes will land ■ Reduced arrival rates – Capacity constraints can cause ripples throughout the National Airspace System (NAS)



Knowledge Check A

REVIEW what you have learned so far about adverse weather. ANSWER the question listed below.

1. Which factors influence aviation weather safety? (Select all correct answers that apply.)

- ☒ Pilot instrument rating
- ☐ Aircraft color
- ☒ Aircraft speed, power, range, and service ceiling
- ☒ Aircraft onboard equipment

Adverse Winds

Adverse wind phenomena include:

Crosswind – When used concerning wind conditions, the word means a wind not parallel to the runway or path of an aircraft.

Aircraft take off and land more efficiently when oriented into the wind. The aircraft's groundspeed is minimized, less runway is required, and the pilot has more time to make adjustments necessary for a smooth landing.

As the wind turns more perpendicular to the runway and becomes a crosswind, aircraft performance gradually degrades and directional control of the aircraft becomes increasingly difficult.

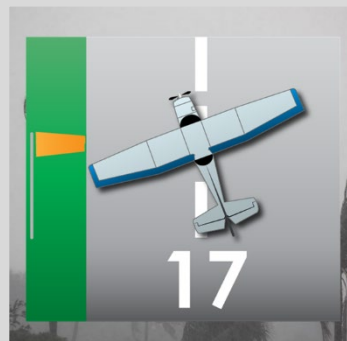
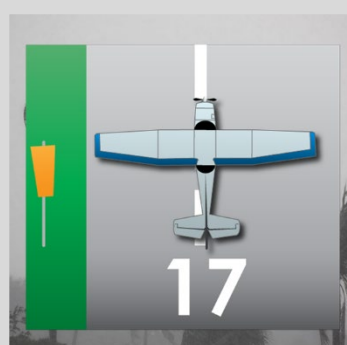
If a pilot does not correctly compensate for the crosswind:

- The aircraft may drift off the side of the runway
- Side load on landing gear can occur, possibly leading to gear collapse

An unforecasted crosswind en route can cause an aircraft to drift off its expected flight path, leading to navigation errors.

Note: All aircraft have a maximum crosswind component at which they can operate.

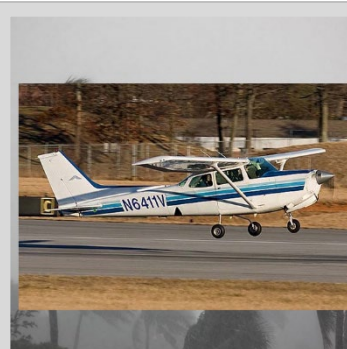
Video – Crosswind Landing Examples (1:01 mins.)



Gust – A sudden, brief increase in the speed of the wind.

Even if the aircraft is oriented into the wind, gusts during takeoff and landing cause airspeed fluctuations which can cause problems for pilots.

- A gust increases airspeed, which may increase lift, and cause an aircraft to briefly rise
- Once the gust ends, a sudden decrease of airspeed occurs, which may decrease lift and cause the aircraft to sink
- This may cause an aircraft to bounce on the runway and possibly lead to a crash



Tailwind – A tailwind is a wind with a component of motion from behind the aircraft.

A tailwind can be hazardous during both takeoff and landing.

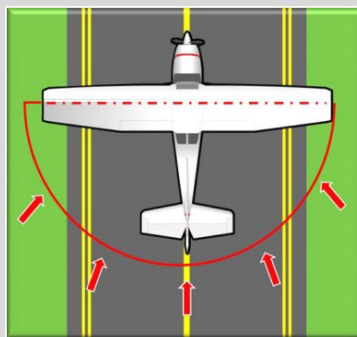
Takeoff

A longer takeoff roll is required. A higher ground speed is required to generate sufficient lift. The aircraft may roll off the end of the runway before liftoff.

A smaller initial rate of climb occurs during takeoff, which may be insufficient to clear obstacles at the end of the runway.

Landing

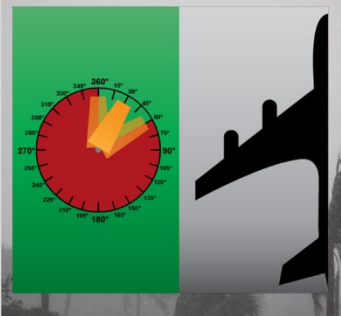
Aircraft experiences a higher ground speed on landing. Thus, the pilot has less time to make adjustments necessary for a smooth landing and may roll off the end of the runway.



Variable Wind – Wind that changes direction frequently.

Wind direction is considered to be variable when, during the 2-minute evaluation period, it fluctuates by 60 degrees or more and the wind speed is more than 6 knots.

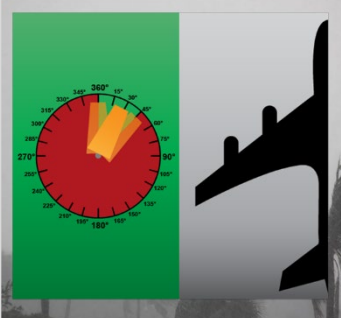
The wind direction may also be considered variable if, during the 2-minute evaluation period, the wind speed is 6 knots or less.



Wind Shift is a change in wind direction of 45 degrees or more which takes place in less than 15 minutes and has sustained winds of 10 knots or more throughout the wind shift.

Variable wind and sudden wind shifts, even at low speeds, can make takeoffs and landings difficult.

- A headwind can quickly become a crosswind or tailwind



Aircraft Size and Wind

Small aircraft are more affected by tailwinds and crosswinds during takeoff and landing than large aircraft. This is because small aircraft generally have slower takeoff and approach speeds than larger aircraft.



Knowledge Check B

REVIEW what you have learned so far about adverse weather. ANSWER the questions listed below.

1. When an aircraft is taking off into a headwind, what will gusts possibly cause it to do? *(Select the correct answer.)*
 - ☐ Drift off the side of the runway
 - ☒ Bounce on the runway
 - ☐ Roll off the end of the runway
2. How can a variable wind be dangerous to an aircraft on takeoff and landing? *(Select the correct answer.)*
 - ☐ Cause engine failure
 - ☒ Quickly become a crosswind or tailwind
 - ☐ Cause the pilot to experience vertigo
3. Which aircraft will perform better in adverse wind conditions due to their higher tailwind and crosswind thresholds? *(Select the correct answer.)*
 - ☐ Smaller aircraft
 - ☐ Helicopters
 - ☒ Larger aircraft

Adverse Weather Summary

Hazardous weather can be devastating to aircraft. Of all the weather phenomenon, adverse wind is the most dangerous when taking off and landing a plane. This lesson identified the severity and impact that weather hazards and adverse wind have on aircraft. Adverse weather and wind direction will make a difference on how air traffic is directed.

INSTRUMENT FLIGHT RULES (IFR) WEATHER

Purpose: This lesson describes the type of weather that requires operation under Instrument Flight Rules and the danger this weather poses to aircraft in this environment.

Objective:

- Describe IFR weather hazards during aircraft takeoff and landing

References for this lesson are as follows:

- Aeronautical Information Manual (AIM)
- FAA-H-8083-28, Aviation Weather Handbook
- American Meteorology Society (AMS), Glossary of Meteorology
- FMH-1, Surface Weather Observations and Reports

Instrument Flight Rules (IFR) Weather

IFR Weather – Weather conditions below the minimum for flight under Visual Flight Rules (VFR).

En route or terminal weather conditions of sufficiently low visibility to require the operation of aircraft under instrument flight rules (IFR).

IFR weather can be hazardous during takeoff and landing.

Continued visual flight into IFR weather is one of the single greatest causes of fatal hazards. VFR → IFR = ☠

Most aircraft accidents related to IFR weather involve pilots who are not instrument qualified.

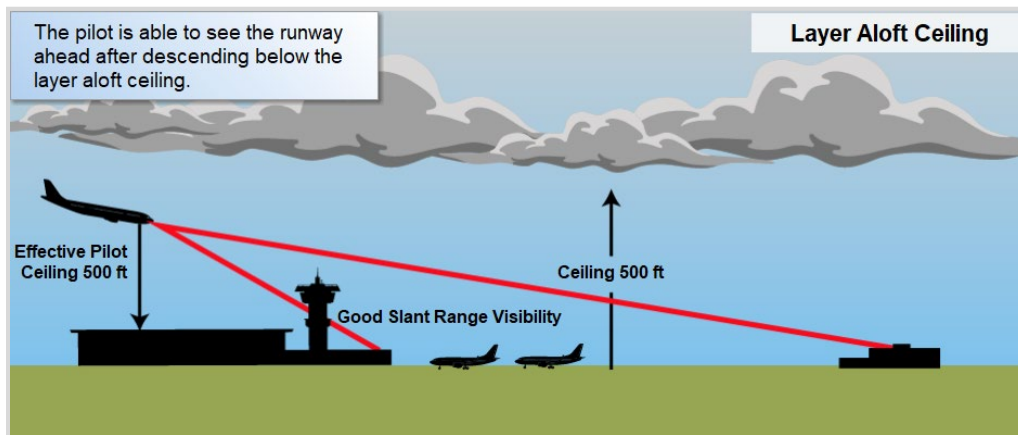
- These pilots attempt flight by visual reference into weather that is suitable only for instrument flight
- The most common cause of these accidents is vertigo (vertigo – the feeling that you or your environment is moving or spinning)
- Pilots also run the risk of flying into unseen obstructions or terrain



Ceiling vs. Indefinite Ceiling

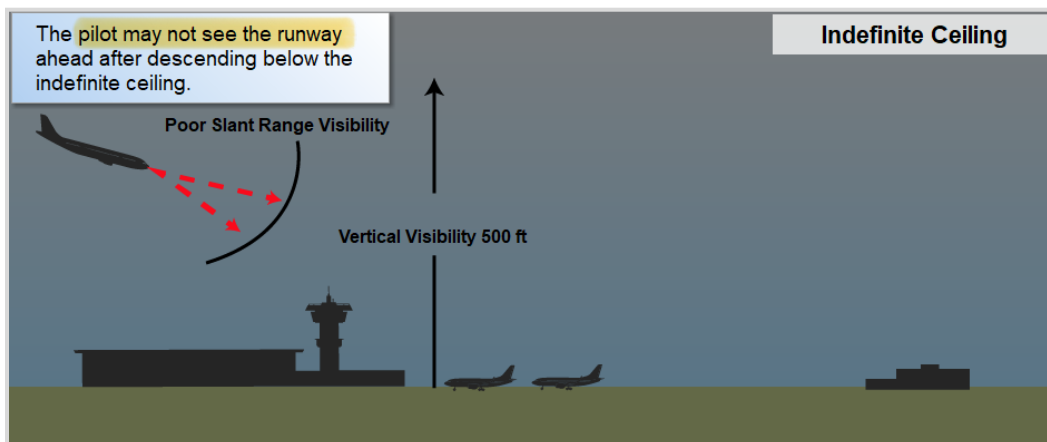
Ceiling – The lowest layer aloft reported as broken or overcast, or the vertical visibility into an indefinite ceiling. There are two distinct types of ceilings, Layer aloft, and indefinite.

Layer Aloft Ceiling – Once a pilot descends below a ceiling caused by a layer aloft or layer aloft ceiling, the pilot can see both the ground below and the runway ahead. Layer aloft includes the lowest broken (BKN) or overcast (OVC) cloud layer. For example, layer aloft will read BKN005, OVC008 on a weather report. An equal ceiling occurs when a layer aloft is unbroken or completely overcast.



Fog

Indefinite Ceiling – The ceiling classification that is applied when the reported ceiling value represents the vertical visibility (VV) upward into a surface-based obscuration. An indefinite ceiling is more hazardous than an equal ceiling caused by a layer aloft; For example, VV005 is more hazardous than BKN005 as shown on a weather report. An indefinite ceiling restricts the pilot's slant-range (air-to-ground) visibility.



Knowledge Check C

REVIEW what you have learned so far about IFR weather. ANSWER the questions listed below.

- Which type of pilot rating is the most at risk for instrument weather aircraft accidents? (Select the correct answer.)
 - ☐ IFR
 - ☒ VFR
 - ☐ VFR and IFR
- What is an indefinite ceiling? (Select the correct answer.)
 - ☒ A surface-based obscuration
 - ☐ Layer aloft obscuration
 - ☐ Overcast cloud layer

Fog

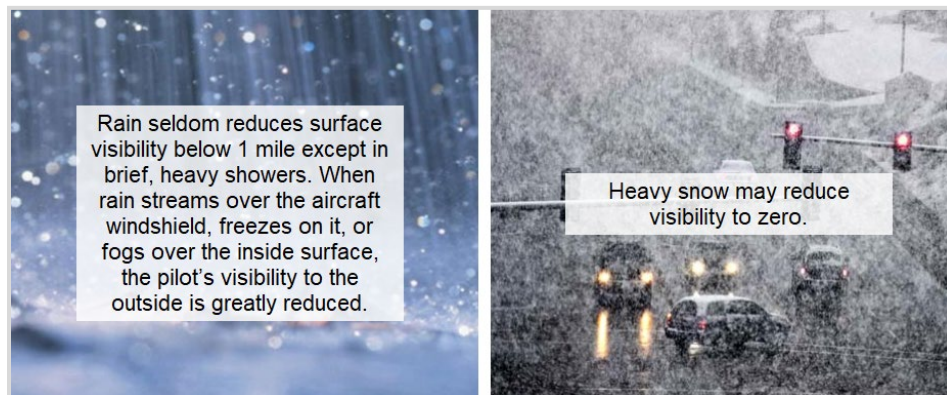
Fog – A visible aggregate of minute water droplets based at the Earth's surface and reducing horizontal visibility to less than 5/8 statute mile. Unlike drizzle, fog does not fall to the ground.

Fog vs. Clouds <i>Surface</i>	Fog differs from clouds only in that its base must be at the Earth's surface while clouds are above the surface.
Fog Formation	Fog forms when the temperature and dew point of the air become identical (or nearly so).
Fog as a Hazard	The speed with which fog can form makes it especially hazardous to aircraft. Fog is one of the most common and persistent weather hazards encountered in aviation.

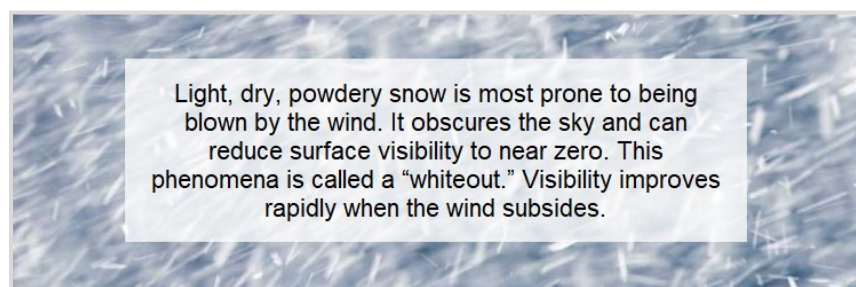


Precipitation

Precipitation – Any of the forms of water particles, whether liquid or solid, that fall from the atmosphere and reach the ground. Rain, drizzle, and snow are the types of precipitation which most commonly produce IFR weather.



Blowing Snow – Snow lifted from the surface of the Earth by the wind to a height of 6 feet or more above the ground and blown about in such quantities that the reported horizontal visibility is reduced to less than 7 statute miles.



Note: Snow falling from clouds is not required to produce blowing snow. In fact, skies above the blowing snow could be clear.

Volcanic Ash Cloud

Volcanic Ash – Fine particles of rock powder that originate from a volcano and that may remain suspended in the atmosphere for long periods.

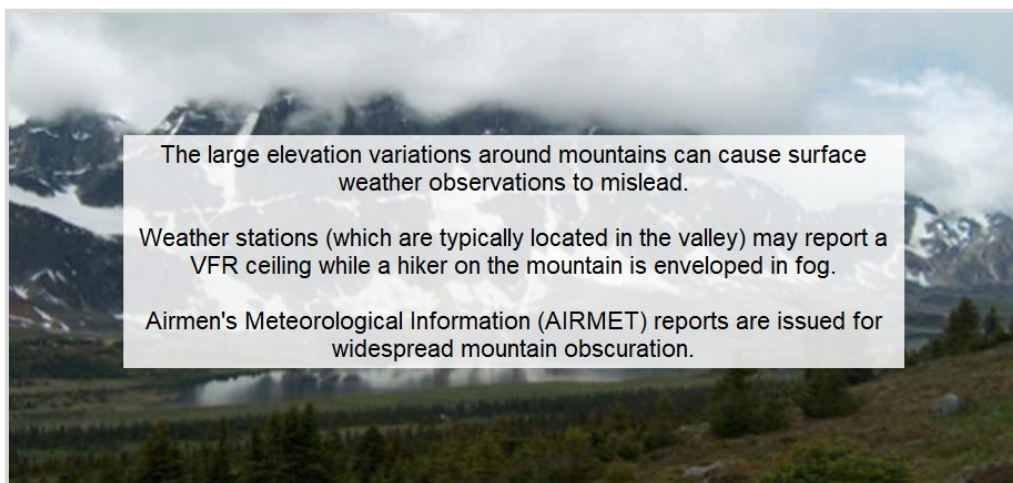
Volcanic Ash vs. Clouds	Even if visible, it is difficult to distinguish visually between an ash cloud and an ordinary cloud. A volcano ash cloud may not be detected by aircraft or Air Traffic Control (ATC) radar.
Volcanic Ash as a Hazard	Aside from producing IFR weather, flying into a volcanic ash plume can be exceedingly dangerous. Ingestion of volcanic ash into an engine can lead to partial or total power loss. Also, ash covering a runway can hide its markings and cause aircraft to lose traction during takeoffs and landings.



Note: If volcanic ash is pulled into a jet engine, it can be heated to temperatures that are higher than the melting temperature of the ash. The ash can melt in the engine and the soft, sticky product can adhere to the inside of the engine. This restricts engine airflow and adds weight. Volcanic ash may not be visible, especially at night or in instrument conditions.

Mountain Obscuration

Mountain Obscuration – Conditions over significant portions of mountainous geographical areas are such that pilots in-flight should not expect to maintain visual meteorological conditions or visual contact with mountains or mountain ridges near their route of flight.





Knowledge Check D

REVIEW what you have learned so far about IFR weather. ANSWER the questions listed below.

1. When does fog form? *(Select the correct answer.)*
 - ☐ Temperature is less than dew point
 - ☒ Temperature equals dew point
 - ☐ Temperature is greater than dew point
2. Which precipitation most commonly produce instrument weather? *(Select the correct answer.)*
 - ☒ Rain, drizzle, and snow
 - ☐ Hail, snow grains, and ice pellets
 - ☐ Snow, snow grains, and ice crystals
3. Which weather phenomenon involves the obscuration of mountain peaks caused by clouds, precipitation, smoke, haze, mist, or fog? *(Select the correct answer.)*
 - ☐ Blowing snow
 - ☐ Volcanic ash cloud
 - ☒ Mountain obscuration

IFR Weather Summary

This lesson described the type of weather that requires operation under IFR and the danger this weather poses to aircraft in this environment. Directing air traffic in instrument weather requires an in-depth knowledge of the weather environment within which pilots operate.

TURBULENCE

Purpose: The purpose of this lesson is to define the environmental conditions that cause aircraft turbulence and the effects they produce. This lesson also describes the levels of turbulence intensity, from light to extreme.

Objectives:

- Identify types of turbulence that affect aircraft
- Identify causes of aircraft turbulence

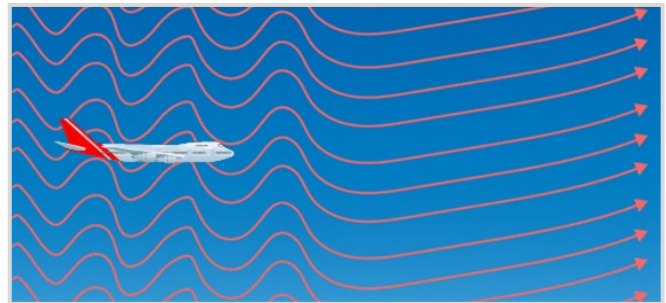
References for this lesson are as follows:

- Aeronautical Information Manual (AIM)
- FAA-H-8083-28, Aviation Weather Handbook
- American Meteorology Society (AMS), Glossary of Meteorology
- UCAR/COMET: Mountain Waves and Downslope Wind

Aircraft Turbulence

Aircraft Turbulence is irregular motion of an aircraft in flight, especially when characterized by rapid up-and-down motion, caused by a rapid variation of atmospheric wind velocities.

Turbulence ranges from annoying bumpiness to severe jolts that cause structural damage to aircraft and/or injury to its passengers.



Causes of Turbulence

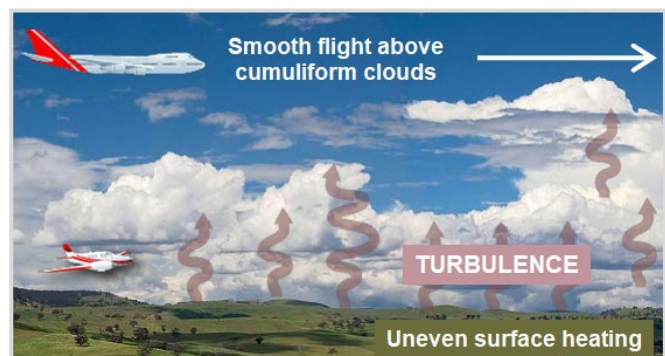
Turbulence is caused by convective currents (called “convective turbulence”), obstructions to wind flow (called “mechanical turbulence”), and wind shear.

Convective Turbulence

Convective turbulence result from convective storms, particularly thunderstorms that are felt by aircraft. The turbulence is caused by strong updrafts and downdrafts.

- Convective currents are most active on warm summer afternoons when winds are light
- Turbulence strength can vary considerably within short distances due to uneven surface heating
 - Barren surfaces and plowed fields become hotter than open water or ground covered by vegetation
- Billowy cumuliform clouds indicate convective turbulence
 - The cloud top marks the upper limit of the convective current
- When the air is too dry for cumuliform clouds to form, convective currents can still be active

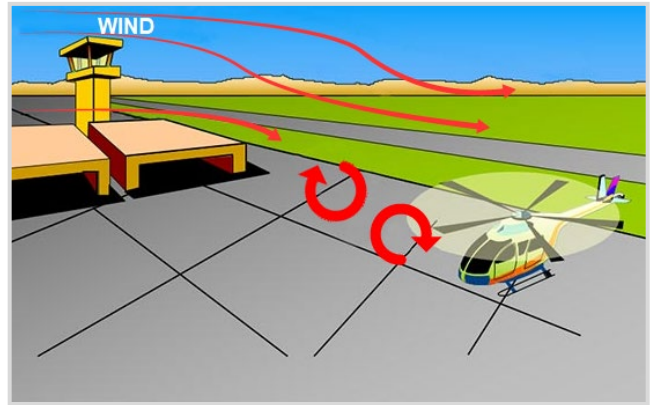
Clouds are NOT necessary for convective turb.



Mechanical Turbulence

Mechanical turbulence is turbulence produced by shear flow. This is caused by obstructions, such as trees, buildings, and mountains.

- Obstructions to the wind flow disrupt the smooth flow of air
- Turbulence intensity is directly related to:
 - Wind speed
 - Shape or roughness of the obstructions

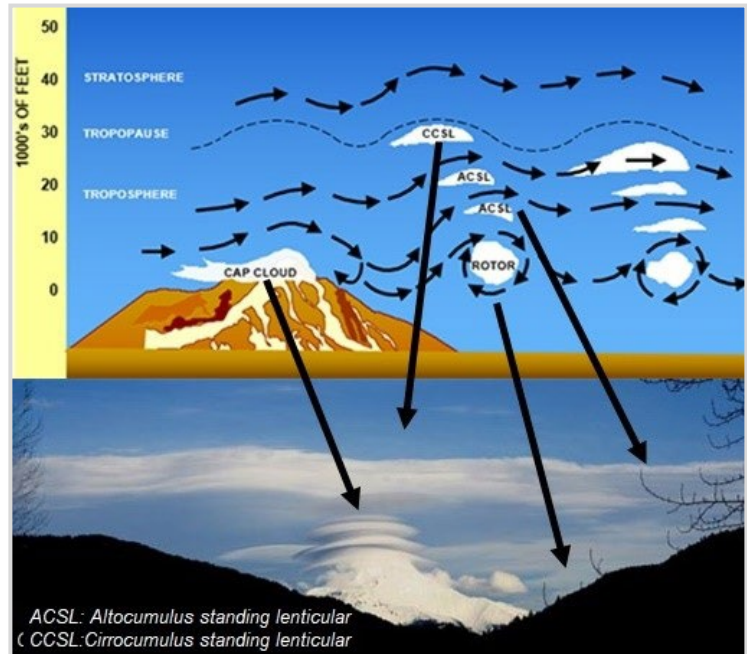


Mountain Waves

A **mountain wave** is an atmospheric gravity wave formed when stable air flow passes over a mountain or mountain barrier.

- Mountain waves develop above and downwind of mountains
 - The waves remain nearly stationary while the wind blows rapidly through them
 - The waves may extend 100 miles or more downwind from the mountain range
 - Mountain waves frequently produce severe to extreme turbulence
 - Location and intensity varies with wave characteristics

Note: Mountain waves usually extend upward into the lower stratosphere.



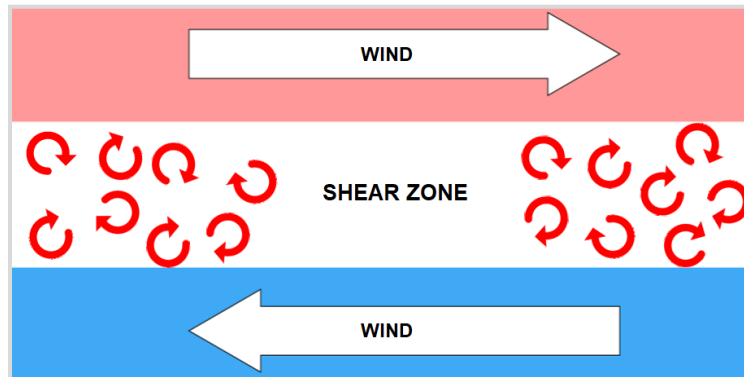
Knowledge Check E

REVIEW what you have learned so far about turbulence. ANSWER the questions listed below.

1. When the air is too dry for cumuliform clouds to form, which type of currents can still cause turbulence? (Select the correct answer.)
 - ☐ Wind shear
 - ☐ Mechanical
 - ☒ Convective
2. Which type of turbulence forms when stable air flows over a mountain? (Select the correct answer.)
 - ☒ Mountain wave
 - ☐ Mountain downwind
 - ☐ Mountain turbulence

Wind Shear Turbulence

Wind Shear is a change in wind speed and/or wind direction in a short distance resulting in a tearing or shearing effect. It can exist in a horizontal or vertical direction and occasionally in both.



Wind Shear Turbulence Conditions

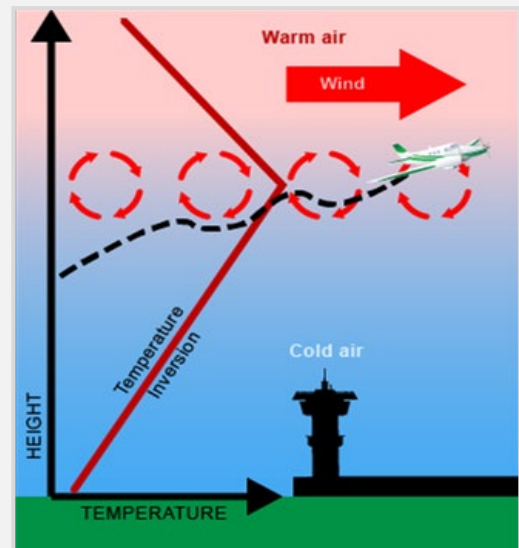
Three conditions of wind shear are of special interest: wind shear with low-level temperature inversion, wind shear with a frontal zone, and clear air turbulence (CAT) associated with the jet stream.

Temperature Inversion

Temperature inversion is a layer in which temperature increases with altitude.

Inversions occur:

- Within the lowest few thousand feet above ground due to nighttime cooling
- Along frontal zones
- When cold air is trapped in a valley

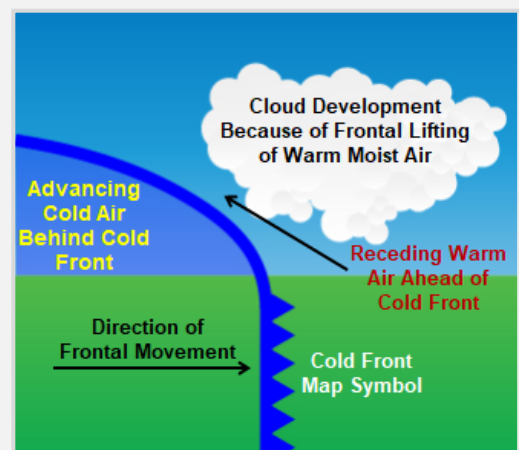


Frontal Zone

Frontal zone is the interface or transition zone between two air masses of different density.

A front can be between two dry air masses and be devoid of clouds but produce strong winds from different directions.

The degree of turbulence depends on the magnitude of the wind shear.

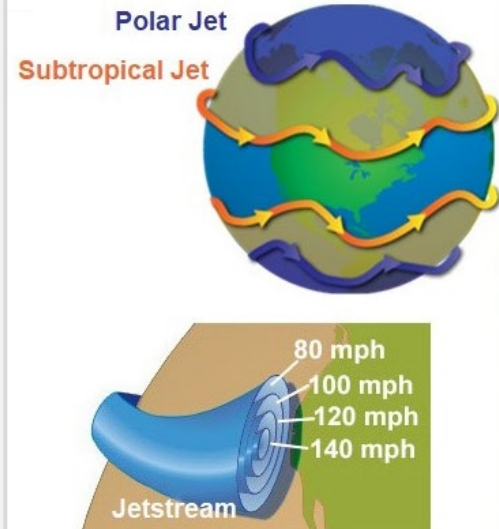


Clear Air Turbulence (CAT)

Clear Air Turbulence (CAT) is a higher altitude (~20,000 to 50,000 ft.) turbulence phenomenon occurring in cloud-free regions associated with wind shear, particularly between the core of a jet stream and the surrounding air.

CAT frequency and intensity are maximized during winter when jet streams are the strongest.

Jet stream CAT is usually best avoided by changing altitude a few thousand feet.



Turbulence Intensity Classifications

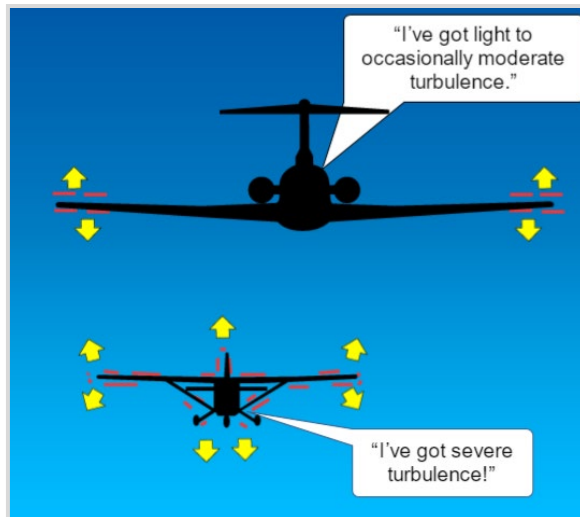
INTENSITY	CODED	AIRCRAFT REACTION	REACTION INSIDE AIRCRAFT
<i>Jeopardy</i> Light	LGT	Momentarily causes slight, erratic changes in altitude and/or attitude (pitch, roll, yaw).	Occupants may feel a slight strain against seat belts or shoulder straps. Unsecured objects may be displaced slightly. Food service may be conducted and little or no difficulty is encountered in walking.
Moderate	MOD	Changes in altitude and/or attitude occur, but the aircraft remains in positive control at all times. It usually causes variations in indicated airspeed.	Occupants feel definite strains against seat belts or shoulder straps. Unsecured objects are dislodged. Food service and walking are difficult.
Severe	SEV	Causes large, abrupt changes in altitude and/or attitude. It usually causes large variations in indicated airspeed. Aircraft may be momentarily out of control.	Occupants are forced violently against seat belts or shoulder straps. Unsecured objects are tossed about. Food service and walking are impossible.
Extreme	EXTRM	Aircraft is violently tossed about and is practically impossible to control. It may cause structural damage.	

Chop is a category of turbulence that causes rapid and somewhat rhythmic bumpiness without appreciable changes in altitude or attitude. May be reported as light chop or moderate chop.

Aircraft Reaction to Turbulence

Aircraft reaction varies with:

- Difference in windspeed in adjacent currents
- Aircraft size
- Wing loading
- Airspeed
- Aircraft altitude



Knowledge Check F

REVIEW what you have learned so far about turbulence. ANSWER the questions listed below.

1. What type of turbulence momentarily causes slight erratic changes in altitude and/or attitude (pitch, roll, yaw)? *(Select the correct answer.)*
 - ☒ Light
 - ☐ Moderate
 - ☐ Severe
2. During what type of turbulence is the aircraft violently tossed about and practically impossible to control? *(Select the correct answer.)*
 - ☐ Moderate
 - ☐ Severe
 - ☒ Extreme
3. What generates turbulence between two wind currents of differing wind directions and/or speeds? *(Select the correct answer.)*
 - ☒ Wind shear
 - ☐ Obstructions to the wind flow
 - ☐ Convective currents

Turbulence Summary

This lesson identified the types and causes of turbulence that affect aircraft, as related to hazardous weather conditions.

HIGH DENSITY ALTITUDE

Purpose: The purpose of this lesson is to describe the effects of high density altitude on aircraft, as related to hazardous weather conditions.

Objective:

- Describe effects of high density altitude on aircraft

Reference for this lesson are as follows:

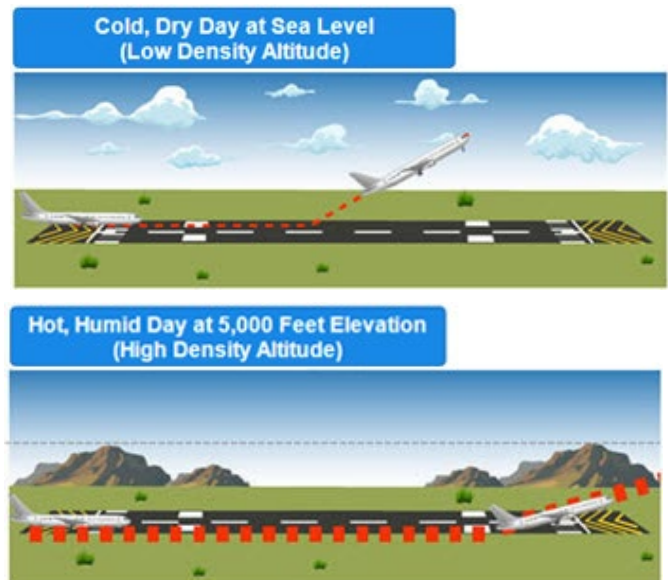
- FAA-H-8083-28, Aviation Weather Handbook
- FAA-P-8740-2, Density Altitude

High Density Altitude

Steps 2 finding the
 $\uparrow$ Density alt. $\Rightarrow$ $\uparrow$ Alt. $\downarrow$ Density

Density altitude is pressure altitude corrected for nonstandard temperature. As the density of the air increases (low density altitude), aircraft performance increases; conversely, as air density decreases (high density altitude), aircraft performance decreases.

- High density altitude can be hazardous during both takeoff and landing
 - It reduces power and thrust
 - A longer takeoff and landing roll are required



Knowledge Check G

REVIEW what you have learned so far about high density altitude. ANSWER the question listed below.

- How does high density altitude affect power and thrust of an aircraft? (Select the correct answer.)
 - ☐ Increases
 - ☒ Reduces
 - ☐ Does not alter

High Density Altitude Summary

This lesson described the effects of high density altitude on aircraft, as related to hazardous weather conditions.

ICING

Purpose: The purpose of this lesson is to define the types of icing hazards for aircraft.

Objectives:

- Identify characteristics of icing
- Identify how icing affects an aircraft

References for this lesson are as follows:

- FAA-H-8083-28, Aviation Weather Handbook
- American Meteorology Society (AMS), Glossary of Meteorology
- UCAR/COMET: Forecasting Aviation Icing: Ice Type and Severity
- AOPA Safety Advisor: Aircraft Icing

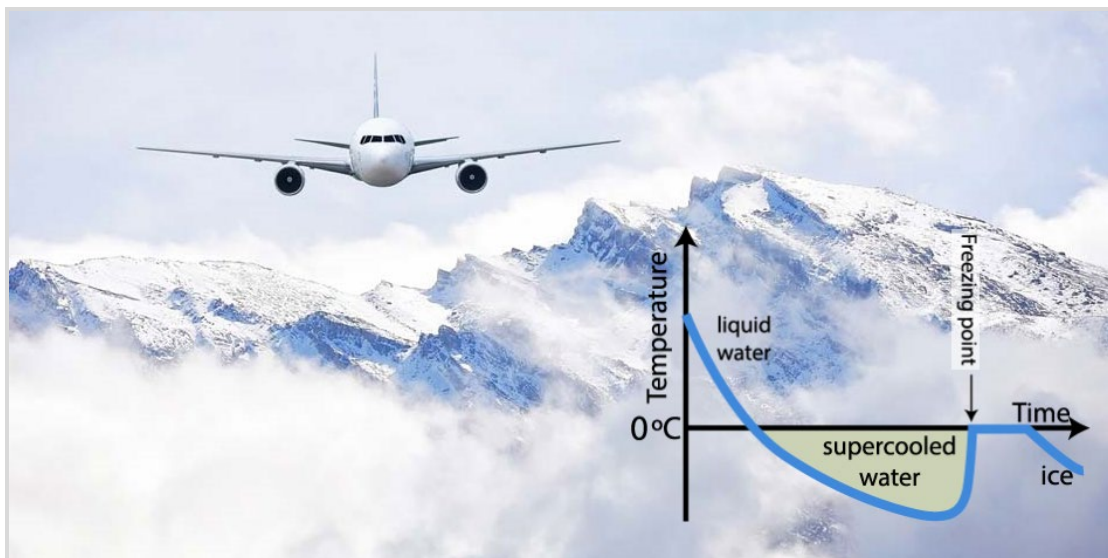
Icing on Aircraft

Icing is any deposit of ice forming on an object. It is a **cumulative hazard** and is one of the major hazards to aviation.



Supercooled Water

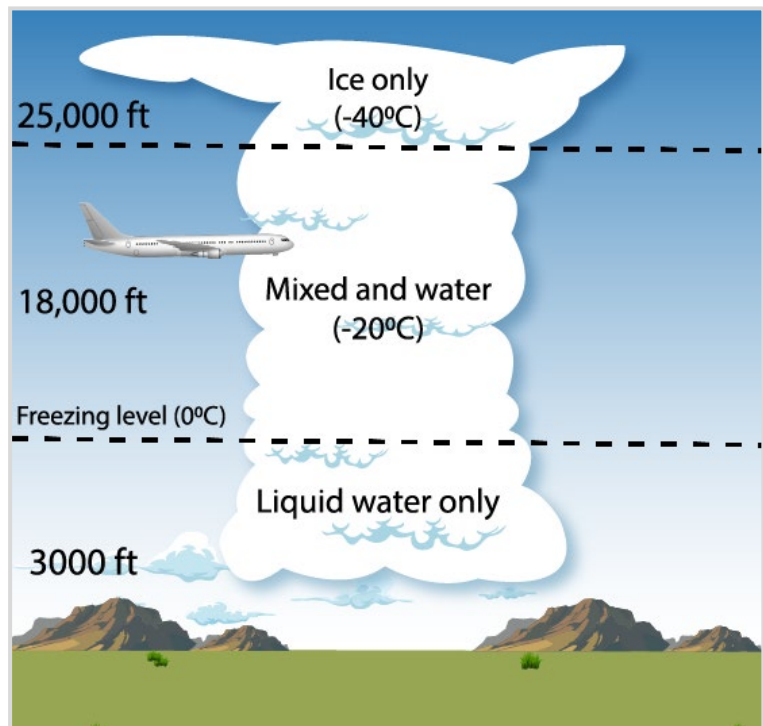
Supercooled water is liquid water at temperatures below the freezing point (0°C). Water chilled below its standard freezing point will crystallize in the presence of a solid around which a crystal structure can form.



Droplets of supercooled water often exist in:

- Stratiform clouds
- Cumulus clouds

Aircraft flying through these clouds see an abrupt crystallization of these droplets, resulting in the formation of ice on the aircraft.



Video – *Principle of Supercooling* (0:40 mins.)

Structural Icing

Structural icing is ice that sticks to the outside of an aircraft. Structural icing forms when supercooled water strikes the aircraft's airframe, and can accumulate on every exposed frontal surface of the aircraft. The three types of structural ice are **rime**, **clear (or glaze)**, and **mixed**.



Rime Ice

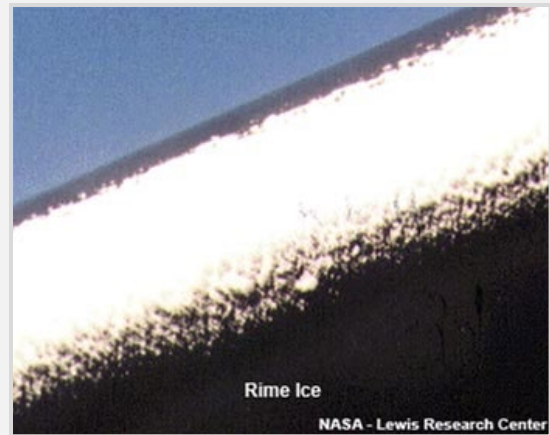
Rime ice is rough, milky, opaque ice formed by the instantaneous freezing of small supercooled water droplets after they strike the aircraft.

Favorable conditions:

- Lower liquid water contents
- Small droplets

Most common, but least serious type:

- Typically controlled by deicers or anti-icers
- Its shape and texture disturb the airflow
- Its brittleness makes it easier to remove



Clear Ice

Clear ice (or glaze ice) is a glossy, clear, or translucent ice formed by the relatively slow freezing of large supercooled water droplets.

Favorable conditions:

- Higher liquid water contents
- Larger droplets

More hazardous than rime:

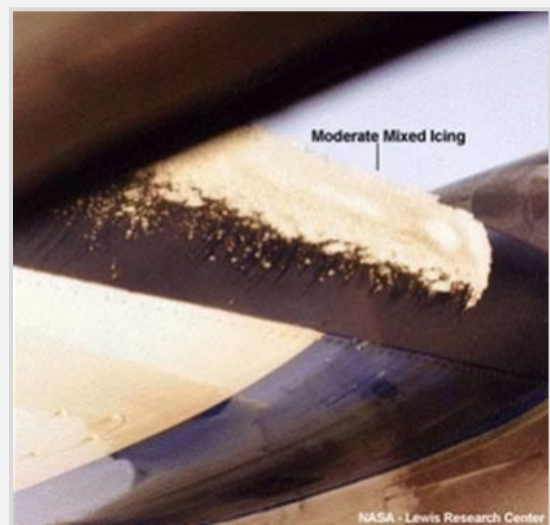
- Can greatly disrupt airflow
- Pilots may not see it
- Can be difficult to remove
- Can spread beyond the reach of deicing/anti-icing equipment



Mixed Ice

Mixed ice is simultaneous appearance or a combination of rime and glaze ice characteristics. Since the clarity, color, and shape of the ice will be a mixture of rime and glaze (clear) characteristics, accurate identification of mixed ice from the cockpit may be difficult.

- Appears as layers of relatively clear and opaque ice when examined from the side
- Mixed icing combines the dangerous effects of both clear and rime icing conditions



Icing Intensity Classification

Icing Intensity	Coded	Description
Trace	TRACE	Ice becomes perceptible. Rate of accumulation slightly greater than sublimation. Deicing/anti-icing equipment is not utilized unless encountered for an extended period of time (over 1 hour).
Light	LGT	The rate of accumulation may create a problem if flight is prolonged in the environment (over 1 hour). Occasional use of deicing/anti-icing equipment removes/prevents accumulation. It does not present a problem if the deicing/anti-icing equipment is used.
Moderate	MOD	The rate of accumulation is such that even short encounters become potentially hazardous and use of deicing/anti-icing equipment or flight diversion is necessary.
Severe	SEV	The rate of accumulation is such that deicing and icing/anti-icing equipment fails to reduce or control the hazard. Immediate flight diversion is necessary.

Factors that Affect Accumulation

- Aircraft type
- Aircraft design
- Altitude
- Airspeed
- Meteorological factors:
 - Supercooled water
 - Temperature
 - Droplet size

Commercial Jets	Small Turboprops
	
Commercial jets are less vulnerable due to: ■ Powerful anti-icing/deicing equipment ■ Tendency to cruise at higher altitudes where temperatures are typically too cold for icing (< -40 degrees Celsius)	Small turboprops are more susceptible to icing due to: ■ Typically flying at lower altitudes where temperatures often support icing (0 to -20 degrees Celsius)

Icing Effects

Structural icing is a significant hazard:

- **Increased drag** while **decreasing** the ability of the airfoil to create **lift**
- Increases weight
 - **Increased weight** is insignificant compared to the airflow disruption
- Can cause antennas to vibrate so severely that they break
- Can block airflow into a pitot tube to cause false airspeed readings

In moderate to severe icing, build up can be significant enough to prohibit continued flight.

- Aircraft may stall at much higher speeds and, for certain aircraft types, lower angles of attack than normal
- Aircraft can roll or pitch uncontrollably and recovery might be impossible



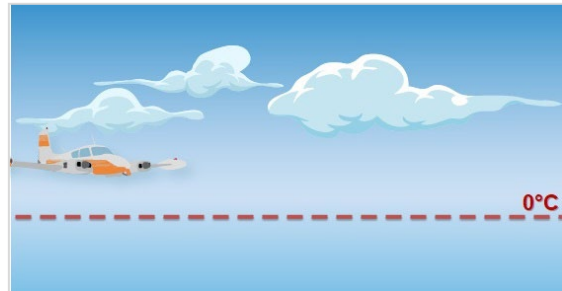
Note: Wind tunnel and flight tests have shown that frost, snow, and ice accumulations (on the leading edge or upper surface of the wing) no thicker or rougher than a piece of coarse sandpaper can reduce lift by 30 percent and increase drag up to 40 percent.

Larger accumulations can reduce lift even more and can increase drag by 80 percent or more.

Avoiding Icing

To escape icing, a pilot can:

- Exit the area of visible moisture
- Climb or descend to positive temperatures
- Climb to altitudes where temperatures are too cold to support supercooled droplets



Knowledge Check H

REVIEW what you have learned so far about icing. ANSWER the questions listed below.

1. What type of icing is the most common and the least serious because it is easier to remove? (Select the correct answer.)
 - ☒ Rime
 - ☐ Clear
 - ☐ Mixed
2. Which level of icing intensity produces a rate of accumulation that may create a problem if the flight is prolonged over 1 hour? (Select the correct answer.)
 - ☐ Moderate
 - ☐ Trace
 - ☒ Light
3. Which of the following are factors that affect accumulation to occur on an aircraft? (Select all correct answers that apply.)
 - ☒ Aircraft type and design
 - ☐ Pilot rating
 - ☒ Meteorological factors

4. Which adverse effect of structural icing is least significant to an aircraft? (*Select the correct answer.*)
- ☐ Increased drag
 - ☒ Increased weight
 - ☐ Decreased lift

Icing Summary

Specific meteorological factors, aircraft type and designs, altitude, and airspeed all contribute to icing in the aviation environment. This lesson Identified the characteristics of icing and how it affects an aircraft in the event you can assist others in need of avoiding or escaping a potentially dangerous icing situation.

THUNDERSTORMS

Purpose: The purpose of this lesson is to describe hazardous effects that occur to aircraft during a thunderstorm.

Objective:

- Identify hazardous effects of thunderstorms on an aircraft

References for this lesson are as follows:

- Aeronautical Information Manual (AIM)
- FAA-H-8083-28, Aviation Weather Handbook
- American Meteorology Society (AMS), Glossary of Meteorology
- NWS: Jetstream: Online School for Weather

Thunderstorms

A **thunderstorm cell** is a storm produced by a cumulonimbus cloud, and always accompanied by lightning and thunder, usually with strong gusts of wind, heavy rain, and sometimes with hail.



Hazardous Effects of Thunderstorms

A thunderstorm can produce almost every aviation weather hazard including:

- Adverse winds
- Instrument Flight Rules (IFR) weather
- Turbulence
- Icing
- Lightning
- Hail
- Downburst/microburst
- Tornado
- Rapid pressure/altimeter changes

A thunderstorm is a storm produced by a **cumulonimbus (CB)** cloud, and is accompanied by some or all:

- Lightning
- Thunder
- Strong gusts of wind
- Heavy rain
- Hail

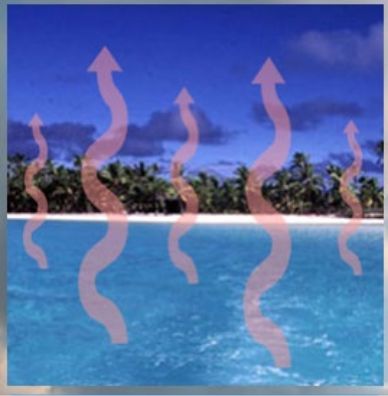
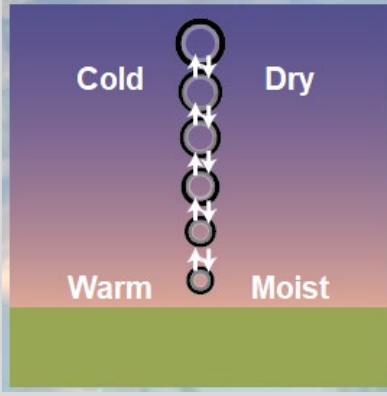
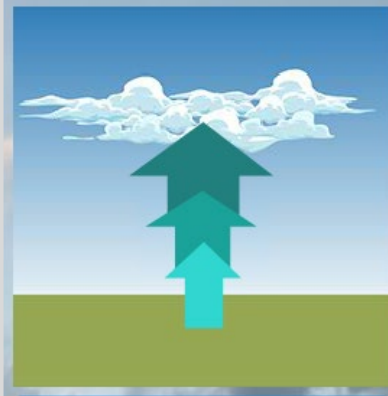


Controllers should anticipate pilot requests for deviation/routing around thunderstorms.

Ingredients for Thunderstorm Cell Formation

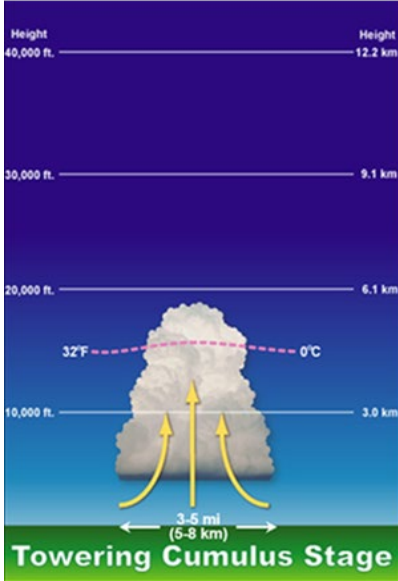
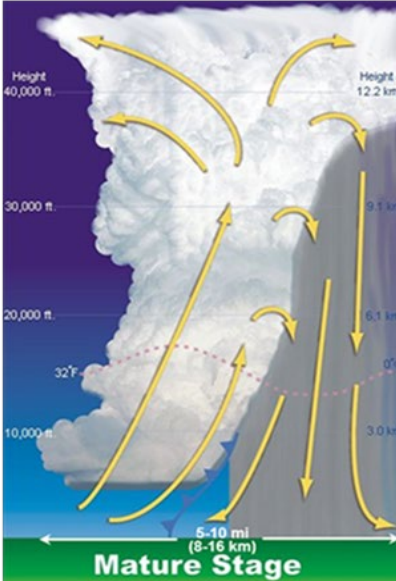
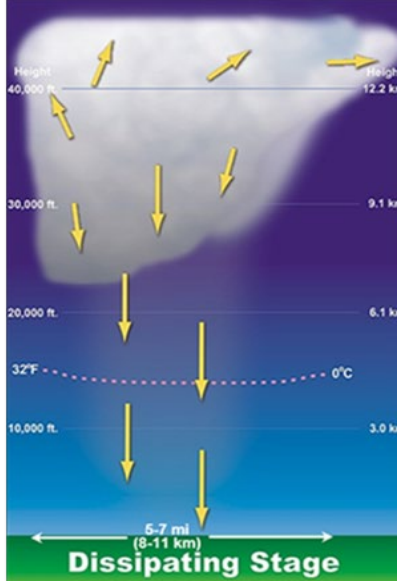
Thunderstorm cell formation requires the following three ingredients:

L.V.A.W

Water Vapor	Unstable Air	Lift
		
Sufficient water vapor ■ Measured using dew point	An atmospheric state warm air below cold air. Since warm air naturally rises above cold air (due to warm air being less dense than cold air), vertical movement and mixing of air layers can occur.	Lifting mechanism(s) strong enough to release the instability ■ Converging winds around surface lows ■ Fronts

Lifecycle Stages of a Thunderstorm Cell

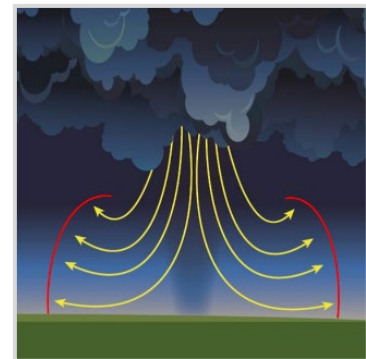
A thunderstorm cell goes through these three distinct stages during its lifecycle, which typically lasts a total of 30 minutes.

Towering Cumulus Stage	Mature Stage	Dissipating Stage
Updrafts can have speeds greater than 3,000 feet per minute, which can exceed the ability of certain aircraft to out climb them	- Precipitation downdraft reaches the surface - Leading edge of downdraft air is called a "gust front" - Weather hazards reach peak intensity	- Precipitation tapers off and ends - Cloud gradually vaporizes from below, leaving only a remnant anvil cloud
 Towering Cumulus Stage	 Mature Stage	 Dissipating Stage

Downburst

A **downburst** is a strong downdraft that induces an outburst of damaging winds on or near the ground.

- Damaging winds are highly divergent
 - Straight
 - Curved
- Sizes of downbursts vary from:
 - ½ mile or less (to)
 - More than 10 miles
- Intense downbursts often cause widespread damage
- Damaging winds
- Downbursts are an extreme hazard to flight, in particular to landing and departing aircraft



Microburst ≤ 2.5 miles in diameter

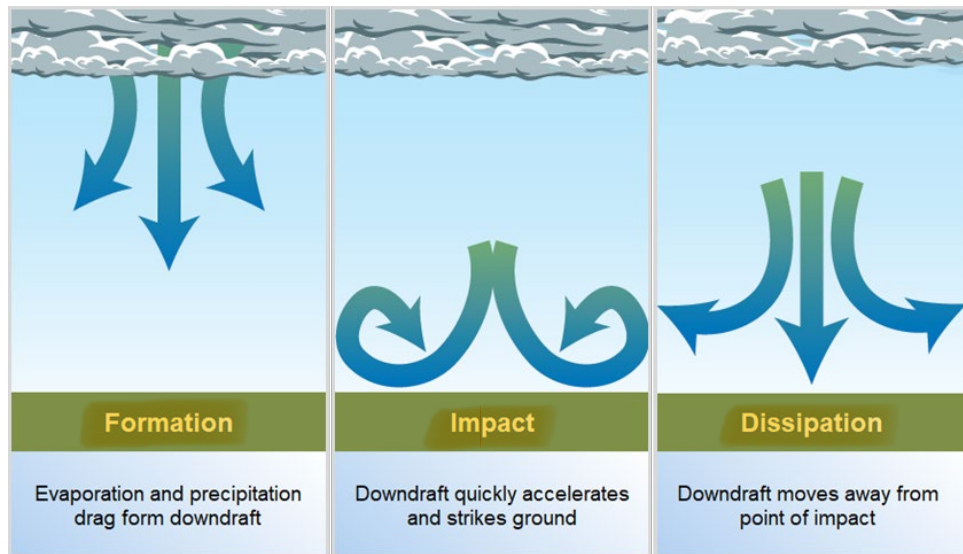
A microburst is a small downburst with outbursts of damaging winds extending 2.5 miles or less. In spite of its small horizontal scale, an intense microburst could induce wind speeds as high as 150 knots.

Macroburst > 2.5 miles in diameter

A macroburst is a convective downdraft with an effected outflow area greater than 2.5 miles wide and peak winds lasting between 5 and 20 minutes. Intense macrobursts may cause tornado-force damage of up to F3 intensity.

Downburst Lifecycle

A strong wind shear associated with a downburst can cause aircraft accidents.



Landing in a Microburst

At point A, the aircraft enters the microburst zone, where a headwind causes it to balloon above the normal glideslope.

At the center of the microburst, **point B**, there is a downdraft, which causes the aircraft to sink.

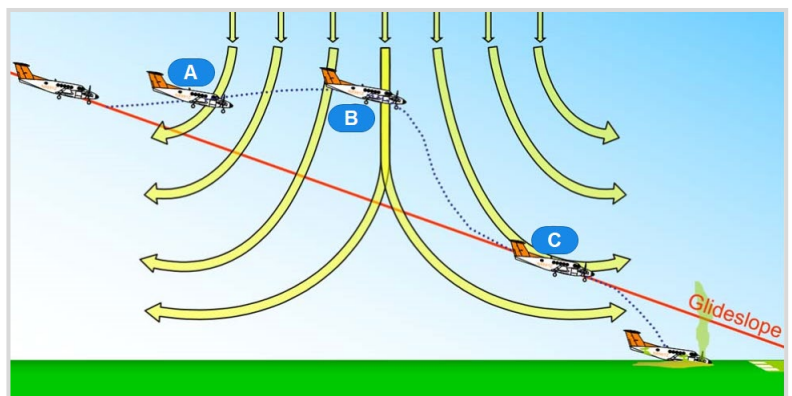
At point C, the aircraft enters the **most lethal zone**, where a sudden tailwind causes the aircraft to lose airspeed and crash.

A downburst is especially dangerous to aircraft when it is encountered **during takeoff or during the approach to landing**.

- During these phases, the aircraft is operating at relatively slow speeds
- A major change of wind velocity can lead to a loss of lift and a crash

Pilots should be alert for a downburst early in the approach phase and be **ready to initiate a missed approach at the first indication**.

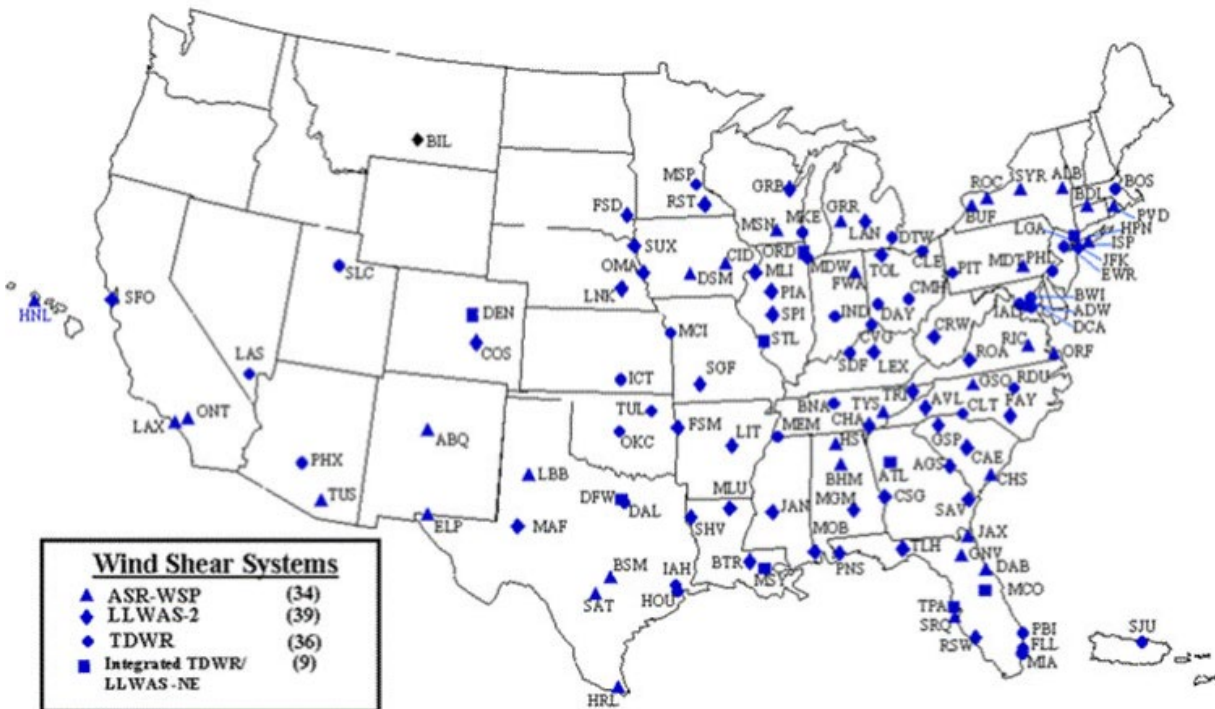
- It may be impossible to recover from a downburst encounter at low altitude

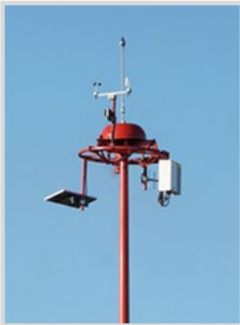
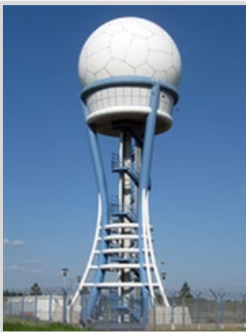


Video – Crash of L-1011 at Dallas-Fort Worth – On August 2, 1985, an L-1011 landing at Dallas-Fort Worth International Airport encountered a microburst that resulted in the deaths of 137 people. This video is a faithful recreation of that tragedy. It provides three views of the final 2 minutes before impact. (6:23 mins.)

Wind Shear Product Systems

To warn terminal controllers and pilots of microbursts, wind shear, and gust fronts, the FAA has installed three wind shear product systems. Tower controllers relay specific alerts from these systems to pilots via voice radio communication.



LLWAS	TDWR	ASR-WSP
		
Low-Level Wind Shear Alert System (LLWAS) detects surface wind shear through the use of up to 32 remote wind sensors situated around an airport.	Terminal Doppler Weather Radar (TDWR) consists of specialized weather radars used to detect microbursts, gust fronts, and convective storms along arrival and departure paths.	Airport Surveillance Radar (ASR)-Weather System Processor (WSP) is an enhanced weather processor for the air traffic control radar that includes Doppler wind estimation for the detection of LLWS.

Video – Vectoring Around Thunderstorms (0:50 mins.)



Knowledge Check I

REVIEW what you have learned so far about thunderstorms. ANSWER the questions listed below.

1. What does a thunderstorm cell formation require besides unstable air and lift? *(Select the correct answer.)*
 - ☐ An inversion
 - ☐ Stratiform clouds
 - ☒ Water vapor
2. Which of the following weather conditions may be impossible to recover from when encountered by an aircraft flying at low altitude? *(Select the correct answer.)*
 - ☐ Headwind
 - ☒ Microburst
 - ☐ Outburst
3. Why is an aircraft in danger when encountering a downburst during takeoff or the approach to landing? *(Select the correct answer.)*
 - ☐ Aircraft is operating at fast speeds, a sudden change in wind direction can throw the aircraft off course
 - ☐ As the aircraft hits the downburst, it causes the aircraft to balloon above the glideslope
 - ☒ Aircraft is operating at relatively slow speeds; if a major change in wind velocity occurs, it can lead to loss of lift and a crash
 - ☐ As the aircraft encounters the downburst, it loses climbing capabilities

Thunderstorms Summary

Knowledge of thunderstorms and the associated hazards is critical to the safety of flight. If conditions are favorable, it may take little time for a thunderstorm to form and can lead to an aircraft accidents or fatalities. As a controller, you should anticipate requests and provide guidance to any aircraft during thunderstorms.

LOW-LEVEL WIND SHEAR (LLWS)

Purpose: The purpose of this lesson is to demonstrate how low-level wind shear affects aircraft during takeoff and landing.

Objective:

- Identify the effects of low-level wind shear (LLWS) on aircraft

References for this lesson are as follows:

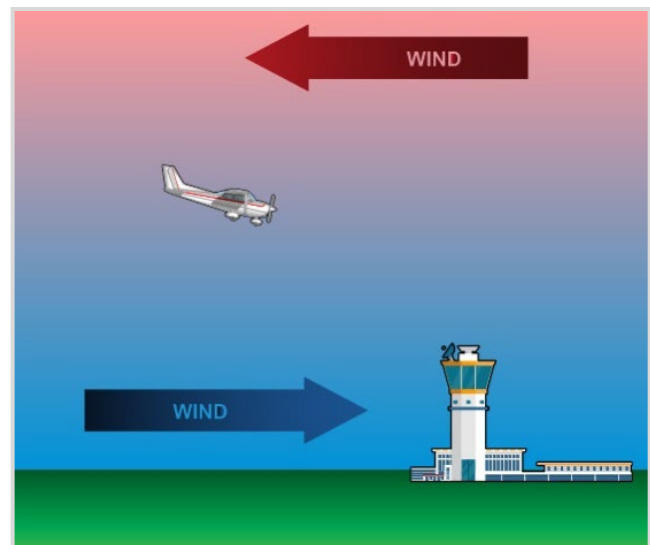
- FAA-H-8083-28, Aviation Weather Handbook
- UCAR/COMET: Writing TAFS For Winds and LLWS

LLWS

Low-level wind shear is a wind shear below 2,000 feet AGL, other than convectively induced, exceeding 10 knots per 100 feet.

LLWS is especially dangerous during takeoff and landing.

- During this phase, aircraft are operating at relatively slow speeds
- A change in wind velocity will cause aircraft to move above or below its glideslope



Wind Shear – Change from Headwind to Tailwind on Landing

While an aircraft is on approach, a shear from a headwind to a tailwind causes:

- Airspeed to decrease
- Nose to pitch down
- Aircraft to drop below the glideslope

If the pilot pulls the nose up to compensate, airspeed will be reduced even further.



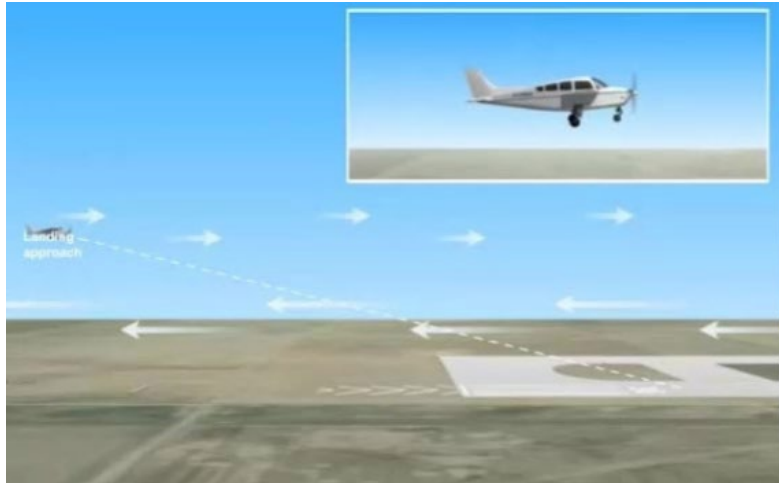
Wind Shear – Change from Tailwind to Headwind on Landing

Conversely, if the wind is calm, or there is a slight tailwind, and the flow shears into a headwind, this causes:

- Airspeed to increase
- Nose to pitch up
- Aircraft to rise upward above the glideslope
- Aircraft to land long and possibly run out of runway

Small, general aviation aircraft are much more prone to the effects of LLWS than large commercial aircraft because their approach speeds are much closer to their stall speeds.

Example: It may only take 15 knots (kt.) of airspeed loss to cause a Cessna 172 to stall, as opposed to greater than 42 kt. for a Boeing 737.



Knowledge Check J

REVIEW what you have learned so far about LLWS. ANSWER the questions listed below.

1. Which of the following effects occur while an aircraft is on approach and encounters a shear from a tailwind to a headwind? (Select all correct answers that apply.)
 - ☒ Increased airspeed
 - ☐ Decreased airspeed
 - ☐ Nose pitches down
 - ☒ Nose pitches up
 - ☒ Aircraft rises upward above glideslope
 - ☐ Aircraft drops below glideslope
2. Why are small general aviation aircraft more prone to the effects of LLWS than large commercial aircraft? (Select the correct answer.)
 - ☐ Aircraft will land long and run out of runway
 - ☐ Pilots compensate by increasing power
 - ☒ Approach speeds are much closer to their stall speed

Low-Level Wind Shear Summary

Controllers must be alert to the effects a low level wind shear could potentially cause small general aviation aircraft. Knowing how an aircraft typically responds when encountering a low level wind shear will allow you to be prepared to react if the aircraft requires assistance.

SUMMARY

This purpose of this module was to identify characteristics of hazardous weather and explain how they affect aviation.

In accordance with Aeronautical Information Manual (AIM); American Meteorology Society (AMS), Glossary of Meteorology; FAA-H-8083-28, Aviation Weather Handbook; NASDAC Review of NTSB Weather Related Accidents; UCAR/COMET: Writing TAFS for Winds and LLWS; Mountain Waves and Downslope Wind; Forecasting Aviation Icing: Ice Type and Severity; AOPA Safety Advisor: Aircraft Icing; FMH-1, Surface Weather Observations and Reports; and NWS: Jetstream: Online School for Weather, you should now be able to:

- Define adverse weather conditions that can affect an aircraft
- Define adverse wind effects that affect aircraft
- Describe IFR weather hazards during aircraft takeoff and landing
- Identify types of turbulence that affect aircraft
- Identify causes of aircraft turbulence
- Describe effects of high density altitude on aircraft
- Identify characteristics of icing
- Identify how icing affects an aircraft
- Identify hazardous effects of thunderstorms on an aircraft
- Identify the effects of low-level wind shear (LLWS) on aircraft

KNOWLEDGE CHECK ANSWER KEY

Knowledge Check A

1. Pilot instrument rating; Aircraft speed, power, range, and service ceiling; Aircraft onboard equipment

Knowledge Check B

1. Bounce on the runway; 2. Quickly become a crosswind or tailwind; 3. Larger aircraft

Knowledge Check C

1. VFR; 2. A surface-based obscuration

Knowledge Check D

1. Temperature equals dew point; 2. Rain, drizzle, and snow; 3. Mountain obscuration

Knowledge Check E

1. Convective; 2. Mountain wave

Knowledge Check F

1. Light; 2. Extreme; 3. Wind shear

Knowledge Check G

1. Reduces

Knowledge Check H

1. Rime 2. Light; 3. Aircraft type and design; Meteorological factors; 4. Increased weight

Knowledge Check I

1. Water vapor; 2. Microburst; 3. Aircraft is operating at relatively slow speeds; if a major change in wind velocity occurs, it can lead to loss of lift and a crash

Knowledge Check J

1. Increased airspeed, Nose pitches up, Aircraft rises upward above glideslope; 3. Approach speeds are much closer to their stall speed